All the Answers are taken from ChatGPT

Q1) Historical Biological Observations: Brownian Motion & Thermodynamics

1. Brownian Motion (5 Marks)

- Discovered by Robert Brown in 1827.
- Observed **random movement of pollen grains** in water under a microscope.
- Movement is due to **continuous collision of water molecules** with particles.
- It proved that:
  - Matter is made of **tiny, constantly moving molecules**.
  - There is **kinetic energy present even in small particles**.
- Later explained using **kinetic theory of matter**.
- Biological significance:
  - Helps in **diffusion of substances** inside cells.
  - Supports **molecular movement in cytoplasm**.

2. Thermodynamics in Biology (5 Marks)

- Study of **energy transformations** in living systems.

First Law of Thermodynamics

- Energy **cannot be created or destroyed**, only transformed.
- In biology:
  - Food energy → **ATP → mechanical/heat energy**.
  - Example: Muscle contraction.

Second Law of Thermodynamics

- Every energy transfer increases **entropy (disorder)**.
- In biology:
  - Cells maintain order but release **heat energy**, increasing disorder in surroundings.
- Important concepts:
  - Living organisms are **open systems** (exchange energy & matter).
  - Metabolism follows thermodynamic laws.

Conclusion (Add for full marks)

- Brownian motion proved **molecular movement**, while thermodynamics explains **energy flow**.
- Both are fundamental to understanding **life processes at cellular level**.

Q2) Classification in Biology Based on Cellularity, Ultrastructure & Nitrogen Excretion

1. Based on Cellularity (3–4 Marks)

- Refers to **number of cells** in an organism.

Types:

- **Unicellular organisms**
    - Made of **single cell**
    - Example: Amoeba, Paramecium
    - One cell performs all functions
- **Multicellular organisms**
    - Made of **many cells**
    - Example: Humans, plants
    - Show **division of labour** (specialized cells)

2. Based on Ultrastructure (3–4 Marks)

- Refers to **internal structure of cells**.

Types:

- **Prokaryotic cells**
    - No **true nucleus** (nuclear membrane absent)
    - No membrane-bound organelles
    - Example: Bacteria
- **Eukaryotic cells**
    - True nucleus present
    - Membrane-bound organelles present
    - Example: Animal cells, Plant cells

3. Based on Nitrogen Excretion (3–4 Marks)

- Refers to **type of nitrogenous waste** excreted.

Types:

- **Ammoniotelic animals**
    - Excrete **ammonia**
    - Highly toxic, needs lots of water
    - Example: Fish
- **Ureotelic animals**

  - Excrete **urea**
  - Less toxic
  - Example: Humans
- **Uricotelic animals**
  - Excrete **uric acid**
  - Least toxic, conserves water
  - Example: Birds, reptiles

Conclusion (for full marks)

- These classifications help in **understanding diversity of life**.
- They are based on **cell structure, organization, and physiological processes**.

Q3) Biomolecules: Chemical Basis of Life

- Biomolecules are **organic compounds** present in living organisms.
- They form the **structure and carry out functions** of cells.
- Major biomolecules: **Carbohydrates, Proteins, Nucleic acids**

1. Carbohydrates (3–4 Marks)

- Composed of **C, H, O** (general formula: $(CH_2O)n$)
- Main function: **energy source**

Types:

- **Monosaccharides** – simple sugars
  Example: Glucose
- **Disaccharides** – two sugar units
  Example: Sucrose
- **Polysaccharides** – complex carbohydrates
  Example: Starch, Glycogen

Functions:

- Provide **instant energy**
- Structural role (cell wall – cellulose)

2. Proteins (3–4 Marks)

- Made of **amino acids** linked by peptide bonds
- Contain **C, H, O, N (sometimes S)**

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Structure levels:

- Primary, Secondary, Tertiary, Quaternary

Functions:

- **Enzymes** (biological catalysts)
  Example: Amylase
- Structural proteins (muscles, tissues)
- Transport (e.g., Haemoglobin)
- Défense (antibodies)

3. Nucleic Acids (3–4 Marks)

- Made of **nucleotides** (sugar + phosphate + nitrogen base)

Types:

- **DNA** (Deoxyribonucleic acid)
  - Stores **genetic information**
- **RNA** (Ribonucleic acid)
  - Helps in **protein synthesis**

Functions:

- Control **heredity and variation**
- Direct **protein formation**

Conclusion (important for full marks)

- Biomolecules are essential for **structure, energy, and regulation** of life.
- They form the **chemical foundation of all living organisms**.

All the Answers are taken from ChatGPT

Q4 Glycolysis (Brief Note)

- Glycolysis means **"splitting of glucose."**
- It is the **first step of cellular respiration**.
- Occurs in the **cytoplasm** of all living cells.
- Takes place in both **aerobic and anaerobic conditions**.

Process of Glycolysis

- One molecule of Glucose (6C) is broken into:
    - Two molecules of Pyruvate (3C each)
- The process involves **10 enzyme-controlled steps**.

Phases of Glycolysis

1. Energy Investment Phase

- Uses **2 ATP molecules**
- Glucose is converted into intermediate compounds

2. Energy Payoff Phase

- Produces:
    - **4 ATP (net gain = 2 ATP)**
    - **2 NADH** (energy carriers)

Net Gain from Glycolysis

- **2 ATP (net)**
- **2 NADH**
- **2 Pyruvate molecules**

Fate of Pyruvate

- **With oxygen (aerobic):**
    - Enters mitochondria → Krebs cycle
- **Without oxygen (anaerobic):**
    - Forms:
        - Lactic acid (in muscles)
        - Ethanol + CO_2 (in yeast)

Importance of Glycolysis

- Provides **quick energy (ATP)**
- First step in **breakdown of glucose**
- Occurs in **all organisms** (universal pathway)

Conclusion (for full marks)

- Glycolysis is a **fundamental metabolic pathway** that converts glucose into pyruvate, producing **energy for cellular activities**.

Q5 Mendel's Laws of Inheritance

- Proposed by Gregor Mendel.
- Based on experiments on Pea plant.
- Explain how traits are **passed from parents to offspring**.

1. Law of Dominance (3–4 Marks)

- When two different alleles are present, **one expresses (dominant)** and the other is **recessive**.

Example:

- Tall (T) × Dwarf (t)
- F_1 generation: All **Tall (Tt)**
  → Tall trait is dominant over dwarf

2. Law of Segregation (3–4 Marks)

- Also called **Law of Purity of Gametes**.
- Alleles **separate during gamete formation**.
- Each gamete gets **only one allele**.

Example:

- Tt × Tt
- Gametes: T and t
- F_2 ratio: **3 Tall : 1 Dwarf**

3. Law of Independent Assortment (3–4 Marks)

- Genes of different traits **assort independently** of each other.

Example:

- Seed shape (Round R / Wrinkled r)
  & Seed colour (Yellow Y / Green y)
- Cross: RrYy × RrYy
- F_2 ratio: **9 : 3 : 3 : 1**

Conclusion (for full marks)

- Mendel's laws form the **basis of genetics**.
- They explain **inheritance patterns and variation** in organisms.

6. Explain DNA as the genetic material with experimental evidence

DNA as the Genetic Material

- Deoxyribonucleic acid (DNA) is the hereditary material in most organisms.
- It carries genetic information from one generation to the next.
- Located mainly in the nucleus (also in mitochondria).

Structure of DNA (2–3 Marks)

- Double helix model proposed by James Watson and Francis Crick.
- Made of nucleotides:
  - Sugar (deoxyribose)
  - Phosphate group
  - Nitrogen bases: A, T, G, C
- Base pairing:
  - A = T, G ≡ C

Experimental Evidence (5–6 Marks)

1. Griffith's Transformation Experiment (1928)

- Conducted by Frederick Griffith.
- Used Streptococcus pneumoniae:
  - S strain (virulent) → kills mice
  - R strain (non-virulent) → harmless
- Observation:
  - Heat-killed S + live R → mice died

- Conclusion:
  - Some "transforming principle" transferred traits from S to R.

2. Avery, MacLeod & McCarty Experiment (1944)

- Scientists: Oswald Avery, Colin MacLeod, Maclyn McCarty.
- Showed that DNA is the transforming principle.
- Method:
  - Destroyed proteins, RNA, or DNA separately
- Result:
  - Transformation stopped only when DNA was destroyed
- Conclusion:
  - DNA is the genetic material

3. Hershey–Chase Experiment (1952)

- Scientists: Alfred Hershey & Martha Chase.
- Used bacteriophages infecting bacteria.
- Method:
  - DNA labelled with radioactive P^{32}
  - Protein labelled with radioactive S^{35}
- Observation:
  - Only DNA entered bacterial cells
- Conclusion:
  - DNA carries genetic information, not protein

Conclusion (for full marks)

- Experiments proved that DNA is the molecule of inheritance.
- It controls structure, function, and reproduction of organisms.

5 Marks

Classification Based on Cellular Organization and Habitat

1. Based on Cellular Organization (2–3 Marks)

- Refers to structure and organization of cells.

Types:

- Unicellular organisms
  - Made of single cell
  - All functions done by one cell
  - Example: Amoeba
- Multicellular organisms
  - Made of many cells
  - Show division of labour
  - Example: Humans
- Also classified as:
  - Prokaryotic (no true nucleus) – Bacteria
  - Eukaryotic (true nucleus present)

2. Based on Habitat (2–3 Marks)

- Refers to natural living environment of organisms.

Types:

- Terrestrial (land)
  - Example: Camel
- Aquatic (water)
  - Example: Fish
- Aerial (air)
  - Example: Eagle
- Amphibious (land & water)
  - Example: Frog

All the Answers are taken from ChatGPT

Classification of Enzymes

- Enzymes are **biological catalysts** that speed up chemical reactions.
- Classified based on the **type of reaction they catalyze**.

Main Classes of Enzymes

1. **Oxidoreductases**
    - Catalyse **oxidation–reduction reactions**
    - Example: Dehydrogenase
2. **Transferases**
    - Transfer **functional groups** (e.g., methyl, amino)
    - Example: Kinase enzymes
3. **Hydrolases**
    - Break bonds using **water (hydrolysis)**
    - Example: Amylase
4. **Lyases**
    - Break or form bonds **without ATP or water**
    - Example: Decarboxylase
5. **Isomerases**
    - Convert molecules into **isomers**
    - Example: Glucose-6-phosphate isomerase
6. **Ligases (Synthetases)**
    - Join two molecules using **ATP**
    - Example: DNA ligase

Conclusion (1 line for full marks)

- Enzyme classification helps in understanding their **specific functions in metabolism**.

🧠 Quick Memory Trick:

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"O T H L I L" → Oxidoreductase, Transferase, Hydrolase, Lyase, Isomerase, Ligase

Concept of Complementation in Human Genetics

- **Complementation** occurs when **two defective genes** from different parents **compensate each other**, restoring a **normal phenotype**.
- It happens when mutations are in **different genes (non-allelic)** controlling the same function.
- Each parent provides a **functional copy of the gene** missing in the other.

Key Points

- If mutations are in **same gene → no complementation**
- If mutations are in **different genes → complementation occurs**
- Helps in identifying whether mutations are in **same or different genes**

Example (Human Genetics)

- **Hereditary deafness**:
  - Two deaf parents may have a **normal (hearing) child**
  - Reason: Mutations are in **different genes** involved in hearing
  - Child gets **one normal allele for each gene**, restoring function

Conclusion (1 line for full marks)

- Complementation explains how **normal traits can appear despite parental defects** when different genes are involved.